

Special Matrices and Transposition

- **Exercise 1**

Let $u = \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}$ with $a, b, c \in \mathbb{R}$. Prove that u is invertible and find u^{-1} .

- **Solution 1.1**

Suppose $u = \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}$ with $a, b, c \in \mathbb{R}$. For any $d, e, f \in \mathbb{R}$, let $A = \begin{bmatrix} 1 & d & e \\ 0 & 1 & f \\ 0 & 0 & 1 \end{bmatrix}$. Then

$$Au = \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & d & e \\ 0 & 1 & f \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & a+d & b+cd+e \\ 0 & 1 & c+f \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

By comparison we have that

$$\begin{cases} a+d=0 \\ c+f=0 \\ b+cd+e=0 \end{cases}.$$

Therefore

$$\begin{cases} d=-a \\ e=ac-b \\ f=-c \end{cases}.$$

And thus

$$A = \begin{bmatrix} 1 & d & e \\ 0 & 1 & f \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -a & ac-b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{bmatrix}.$$

Check the answer by $uA = I$, that is

$$uA = \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -a & ac-b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I.$$

Therefore u is invertible and $u^{-1} = \begin{bmatrix} 1 & -a & ac-b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{bmatrix}.$

- **Exercise 2**

Write $A = \begin{bmatrix} 4 & 5 \\ 1 & 3 \end{bmatrix}$ as the sum of a symmetric matrix and an antisymmetric matrix.

- **Solution 2.1**

Suppose $A \in M_{2 \times 2}(\mathbb{R})$, $A = \frac{A + A^\top}{2} + \frac{A - A^\top}{2}$. We have that $A^\top = \begin{bmatrix} 4 & 1 \\ 5 & 3 \end{bmatrix}$ and thus

$$\begin{aligned} \frac{A + A^\top}{2} &= \begin{bmatrix} 4 & 3 \\ 3 & 3 \end{bmatrix} \\ \frac{A - A^\top}{2} &= \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} \end{aligned}$$

since $\frac{A + A^\top}{2}$ is symmetric and $\frac{A - A^\top}{2}$ is antisymmetric.

• **Exercise 3**

Let $D = \text{diag}(1, 2, 4)$ and let $A = [a_{ij}] \in M_{3 \times 3}(\mathbb{R})$. Determine all matrices satisfying $AD = DA$.

◦ **Solution 3.1**

We are looking for matrices

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

such that $AD = DA$. Then we have

$$\begin{aligned} AD &= \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix} = \begin{bmatrix} a_{11} & 2a_{12} & 4a_{13} \\ a_{21} & 2a_{22} & 4a_{23} \\ a_{31} & 2a_{32} & 4a_{33} \end{bmatrix} \\ DA &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 2a_{21} & 2a_{22} & 2a_{23} \\ 4a_{31} & 4a_{32} & 4a_{33} \end{bmatrix} \end{aligned}$$

By comparison, we have that $a_{11}, a_{22}, a_{33} \in \mathbb{R}$ and $a_{12} = a_{13} = a_{21} = a_{23} = a_{31} = a_{32} = 0$. Thus A is diagonal.

• **Exercise 4**

Suppose $A \in M_{n \times n}(\mathbb{R})$ is both upper triangular and symmetric. Prove that A must be diagonal. Is the converse true?

◦ **Proof 4.1**

Suppose that $A \in M_{n \times n}(\mathbb{R})$ is both upper triangular and symmetric. Then by definition $\forall i > j, a_{ij} = 0$ and $\forall i, j, a_{ij} = a_{ji}$. Now we need to prove that $\forall i, j, i < j, a_{ij} = 0$. As A is symmetric $a_{ij} = a_{ji} = 0$. Thus $\forall i, j, i \neq j, a_{ij} = 0$ which is the definition of diagonal.

The converse is true. Suppose A is diagonal, $A \in M_n(\mathbb{R})$. By definition we have that $\forall i, j, i \neq j, a_{ij} = 0$. Then $\forall i, j, i > j, a_{ij} = 0$ and then A is upper triangular. If $i \neq j$, we have that $a_{ij} = a_{ji} = 0$ and thus A is symmetric.

• **Exercise 5**

Suppose $A \in M_{n \times n}(\mathbb{R})$ is invertible and symmetric. Prove using definition of inverse and identities of transpose that A^{-1} is also symmetric.

◦ **Proof 5.1**

Suppose $A \in M_{n \times n}(\mathbb{R})$ is invertible and symmetric. Then $\exists A^{-1} \in M_n(\mathbb{R}), AA^{-1} = A^{-1}A = I$ and $A = A^\top$. We want to prove $(A^{-1})^\top = A^{-1}$. That is, $(A^{-1})^\top$ is the inverse of A . We have that

$$I = I^\top = (AA^{-1})^\top = (A^{-1})^\top A^\top = (A^{-1})^\top A = I.$$

By the uniqueness of inverse, $(A^{-1})^\top$ is the inverse of A .